

Assignment DM finished

Saturday, 8 March 2025 18:41

2.1

7. The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K -nearest neighbors.

- Compute the Euclidean distance between each observation and the test point, $X_1 = X_2 = X_3 = 0$.
- What is our prediction with $K = 1$? Why?
- What is our prediction with $K = 3$? Why?

a) Euclidean distance

$$d(X_i, X_j) = \sqrt{\sum_{k=1}^K (x_k - x_j)^2}$$

$$d_{obs1} = \sqrt{(0-0)^2 + (3-0)^2 + (0-0)^2} = 3$$

$$d_{obs2} = \sqrt{4} = 2$$

$$d_{obs3} = \sqrt{1+9} \approx 3.16$$

$$d_{obs4} = \sqrt{5} \approx 2.24$$

$$d_{obs5} = \sqrt{2} \approx 1.41$$

$$d_{obs6} = \sqrt{3} \approx 1.73$$

b) With $K=1$ we consider the smallest observations, in this case obs_5 because the distance is 1 the prediction will be classified as 'Green'

c) For $K=3$, the smallest distance can be found in obs_5 and also obs_3 . So the prediction will be 'Red'

OBS #	d	Y
5	1.41	GREEN
6	1.73	RED
2	2	RED
4	2.24	GREEN
1	3	RED
3	3.16	RED

$\left. \begin{matrix} k=1 \\ k=3 \end{matrix} \right\}$

2.2

3. Suppose we have a data set with five predictors, $X_1 = \text{GPA}$, $X_2 = \text{IQ}$, $X_3 = \text{Level}$ (1 for College and 0 for High School), $X_4 = \text{Interaction between GPA and IQ}$, and $X_5 = \text{Interaction between GPA and Level}$. The response is starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model, and get $\beta_0 = 50, \beta_1 = 20, \beta_2 = 0.07, \beta_3 = 35, \beta_4 = 0.01, \beta_5 = -10$.

- Which answer is correct, and why?
 - For a fixed value of IQ and GPA, high school graduates earn more, on average, than college graduates.
 - For a fixed value of IQ and GPA, college graduates earn more, on average, than high school graduates.

iii. For a fixed value of IQ and GPA, high school graduates earn more, on average, than college graduates provided that the GPA is high enough.

iv. For a fixed value of IQ and GPA, college graduates earn more, on average, than high school graduates provided that the GPA is high enough.

- Predict the salary of a college graduate with IQ of 110 and a GPA of 4.0.

- True or false: Since the coefficient for the GPA/IQ interaction term is very small, there is very little evidence of an interaction effect. Justify your answer.

$$X_1 = \text{GPA}; X_2 = \text{IQ}; X_3 = \text{Level} \left(\begin{matrix} 1 = \text{College} \\ 0 = \text{HS} \end{matrix} \right); X_4 = \text{GPA} \times \text{IQ}; X_5 = \text{GPA} \times \text{LEVEL}$$

$Y = \text{starting salary after graduation}$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5$$

$$= 50 + 20X_1 + 0.07X_2 + 35X_3 + 0.01X_4 - 10X_5$$

a) As GPA is also present in X_5 inside an interaction term we could tell that if GPA is not high enough this coefficient term could penalize the whole expression.

So, (iv)

$$\begin{aligned} Y_{col} - Y_{hs} &= (50 + 20X_1 + 0.07X_2 + 35 \cdot 1 + 0.01X_4 - 10X_5) - (50 + 20X_1 + 0.07X_2 + 35 \cdot 0 + 0.01X_4 - 10X_5) \\ &= (35 + 20X_1 + 0.07X_2 + 0.01X_4 - 10X_5) - (50 + 20X_1 + 0.07X_2 + 0.01X_4) \\ &= (35 - 10X_5) - (50 + 20X_1) \\ &= 35 - 10X_5 \\ X_5 &= 35/10 \approx 3.5 \quad \text{threshold} \end{aligned}$$

b) $Y = 50 + (20 \cdot 4) + (0.07 \cdot 110) + (35 \cdot 1) + 0.01(4 \cdot 110) - 10 \cdot (4 \cdot 1) \approx 137.1$

c) False. To understand the interaction effect one needs to calculate if it is significant, it happens when $p\text{-value} < 0.05$

2.3

6. Suppose we collect data for a group of students in a statistics class with variables $X_1 = \text{hours studied}$, $X_2 = \text{undergrad GPA}$, and $Y = \text{receive an A}$. We fit a logistic regression and produce estimated coefficient, $\beta_0 = -6, \beta_1 = 0.05, \beta_2 = 1$.

- Estimate the probability that a student who studies for 40h and has an undergrad GPA of 3.5 gets an A in the class.
- How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

a) Logistic regression $\frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2}}$

$$P(Y=1) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2}} = \frac{e^{-6 + 0.05 \cdot 40 + 1 \cdot 3.5}}{1 + e^{-}} \approx 37.7\%$$

b) X_1 ?

$$P(Y) = 50\%$$

$$0.5 = \frac{1}{1 + e^{-z}}$$

$$0.5 \cdot (1 + e^{-z}) = 1$$

$$0.5 + 0.5e^{-z} = 1$$

$$0.5e^{-z} = 1 - 0.5 = 0.5$$

$$\frac{0.5e^{-z}}{0.5} = \frac{0.5}{0.5}$$

$$e^{-z} = 1$$

$$\ln(e^{-z}) = \ln(1)$$

$$-z = 0$$

$$z = 0$$

$$\text{So, since } z = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

$$0 = -6 + (0.05X_1) + (1 \cdot 3.5)$$

$$6 - 3.5 = 0.05X_1$$

$$X_1 = 2.5/0.05 = 50 \quad \text{hours to study to get an A}$$

2.4

2.4 A retailer wants for marketing purposes distinguish between costumers younger then 35 and customers older then 35. The following table summarizes the data set in the data base of the retailer in an abstract form. The relevant attributes, determined by domain knowledge, are for convenience denoted by A and B . The values for A are $a1, a2$ and $a3$. The values for B are $b1$ and $b2$. The retailer wants to use Data Mining

A	B	Number of Instances	
		Y	O
a1	b1	4	10
a2	b1	6	2
a3	b1	8	6
a1	b2	2	2
a2	b2	6	8

techniques to classify the costumers in the class "young", denoted by Y , and "old", denoted by O .

- Assume a new customer enters the web-store and the retailer has no information at all about this customer. How will this new customer be classified based on the above data and explain why.
- Now assume a new customer comes in for which the retailer knows the values for attribute $A = a3$ and $B = b2$. Is it possible to apply the standard (non Naive) Bayes to classify this new customer? Explain what the problems is.
- Hence the retailer decides to use a naive Bayes classifier for the classification of this new customer.
 - How will this customer now be classified based on the values of A and B ? Explain your answer.

a) If a new customer enters the web-store and we don't have any information about him, we can use the prior probabilities to classify the customer:

$$\text{Total number of instances} = 54$$

$$P(Y) = \frac{26}{54} \approx 0.48$$

$$P(O) = \frac{28}{54} \approx 0.52$$

Since $P(O) > P(Y)$, the new customer will be classified as old "O".

b) $A = a3, B = b2$

The standard Bayes Theorem requires calculating the posterior probabilities of each class, given the known attributes.

$$P(Y|A=a3, B=b2) = \frac{P(A=a3, B=b2|Y) \cdot P(Y)}{P(A=a3, B=b2)}$$

$$P(O|A=a3, B=b2) = \frac{P(A=a3, B=b2|O) \cdot P(O)}{P(A=a3, B=b2)}$$

Considering the given dataset, there is no data for $(A=a3, B=b2)$.

so we cannot apply standard Bayes Theorem because of zero-frequency issue and no prior knowledge

c) If we assume independence of A and B (Naive Bayes), we can estimate the probabilities separately.

$$P(Y|A=a3, B=b2) = \frac{P(A=a3|Y) \cdot P(B=b2|Y) \cdot P(Y)}{P(A=a3) \cdot P(B=b2)} = \left(\frac{8}{26}\right) \cdot \left(\frac{2}{26}\right) \cdot \left(\frac{26}{54}\right) \approx 0.31 \cdot 0.31 \cdot 0.48 \approx 0.046$$

$$P(O|A=a3, B=b2) = \frac{P(A=a3|O) \cdot P(B=b2|O) \cdot P(O)}{P(A=a3) \cdot P(B=b2)} = \left(\frac{6}{28}\right) \cdot \left(\frac{8}{28}\right) \cdot \left(\frac{28}{54}\right) \approx 0.21 \cdot 0.36 \cdot 0.52 \approx 0.039$$

Considering that the denominator is the same, we can just compare the numerators.

Since $P(Y|A=a3, B=b2) > P(O|A=a3, B=b2)$, the new customer will be classified as Young "Y".

2.5

2.5 Consider the same dataset as in the previous question. Now the retailer (data analyst) wants to use Decision Trees to classify new customers.

- What is the classification error rate for attribute A ?
- What is the classification error rate for attribute B ?
- What will be the splitting attribute in the top (root) of the Decision Tree if one uses the classification error rate?
- What is the Gini index for attribute A ?
- What is the Gini index for attribute B ?
- What will be the splitting attribute in the top (root) of the Decision Tree if one uses the Gini index?
- Construct the full Decision Tree, using the error rate as heuristic, and what is the overall classification error rate on the above dataset?
- Is this classification error rate an optimistic or pessimistic estimate of the error rate on unseen new data? Explain your answer.

a) $\text{Classification Error Rate}(A) = \frac{6+4+6}{54} \approx 0.30$

A	Y	O	label	# errors	TOT
a1	6	18	0	6	24
a2	12	4	1	4	16
a3	8	6	1	6	14

b) $\text{Classification Error Rate}(B) = \frac{8+18}{54} \approx 0.481$

A	Y	O	label	# errors	TOT
b1	18	18	1	18	36
b2	8	10	0	8	18

c) If one uses classification error rate as a measure, the splitting attribute in the top (root) of the DT will be "A" because we seek for a smaller class error value to minimize entropy.

d) $\text{Gini} = \sum_{i=1}^L \hat{p}_{im} (1 - \hat{p}_{im}) = 1 - \sum_{i=1}^L p_i^2$ where p_i = proportion of instances in class i
 $= \frac{\sum \text{category total}}{\text{total samples}} \cdot 6 (\text{categories})$

$$G(A) = \frac{24}{54} \cdot 6 + \frac{16}{54} \cdot 6 + \frac{14}{54} \cdot 6 = 0.43$$

$$G_{a1} = 1 - \left[\left(\frac{6}{24}\right)^2 + \left(\frac{18}{24}\right)^2 \right] = \frac{7}{8} = 0.75$$

$$G_{a2} = 1 - \left[\left(\frac{12}{16}\right)^2 + \left(\frac{4}{16}\right)^2 \right] = \frac{3}{8} = 0.375$$

$$G_{a3} = 1 - \left[\left(\frac{8}{14}\right)^2 + \left(\frac{6}{14}\right)^2 \right] = \frac{24}{49} \approx 0.49$$

$$G(B) = \frac{24}{54} \cdot 0.375 + \frac{18}{54} \cdot 0.375 + \frac{14}{54} \cdot 0.49 \approx 0.405$$

e) $G(B) = \frac{36}{54} \cdot 6 + \frac{18}{54} \cdot 6 = 0.3$

$$G_{b1} = 1 - \left[\left(\frac{18}{36}\right)^2 + \left(\frac{18}{36}\right)^2 \right] = \frac{1}{2} = 0.5$$

$$G_{b2} = 1 - \left[\left(\frac{8}{18}\right)^2 + \left(\frac{10}{18}\right)^2 \right] = \frac{40}{81} \approx 0.49$$

$$G(B) = \frac{36}{54} \cdot 0.5 + \frac{18}{54} \cdot 0.49 = 0.498$$

f) Same as classification error rate, we look for the smaller value, the root of the DT will be "A".

g) From the previous answers, we know that the root is A so we split A into his 3 branches. Then, we calculate the error rates:

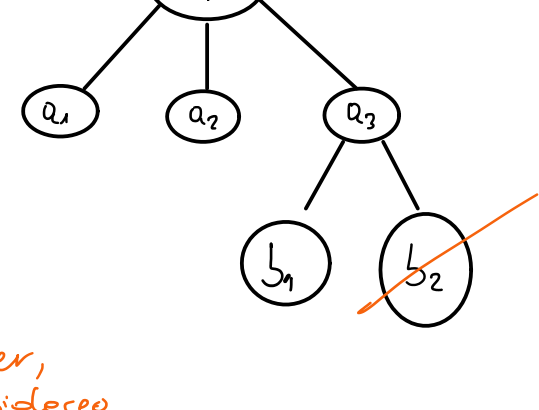
$$\text{ErrorRate} = \frac{\text{category errors}}{\text{category total}} = \frac{\text{misclassified instances}}{\text{tot. instances}}$$

$$\text{ErrorRate}(a_1) = \frac{6}{24} \approx 0.25$$

$$\text{ErrorRate}(a_2) = \frac{4}{16} \approx 0.25$$

$$\text{ErrorRate}(a_3) = \frac{6}{14} \approx 0.43$$

Since this is the higher, further splits are considered



Since $a_3 > a_1, a_2$, we decide that $a_1 \rightarrow 0, a_2 \rightarrow 1$ and a_3 will be split into b_1 and b_2 but we only have b_1 .

$$\text{ErrorRate}(a_3, b_1) = \frac{6}{14} = 0.43 \rightarrow \text{changed to 'Y' based on the majority class}$$

$$\text{Overall Classification Error Rate} = \frac{6+4}{54} \approx 0.203$$

h) Since the accuracy is $1 - 0.203 \approx 0.80$ we can say it is an optimistic estimate

Assignment Data Mining

2.6 DATASET: VOOBEELD7

(a) ScatterPlot

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels as stm
import pyreadstat
```

```
choll, meta = pyreadstat.read_sav("./voorbeeld7_1-1.sav")
print(meta.column_names)
choll.head()
```

```
['id', 'chol', 'leeftijd', 'bmi', 'actief', 'roken', 'sekse',
 'alcohol']
```

	id	chol	leeftijd	bmi	actief	roken	sekse	alcohol
0	1.0	7.9	66.0	27.41	1.0	0.0	1.0	0.0
1	2.0	7.5	83.0	26.85	1.0	0.0	1.0	2.0
2	3.0	7.6	84.0	27.24	1.0	0.0	1.0	2.0
3	4.0	6.0	65.0	27.50	1.0	0.0	1.0	2.0
4	5.0	6.2	56.0	27.78	1.0	0.0	1.0	0.0

```
print(choll.describe())
# sekse=sex is a factorial variable
choll.alcohol.unique() #also alcohol so i will convert them to factors
##choll = pd.get_dummies(choll, columns=["sekse", "alcohol"],
drop_first=True)
```

	id	chol	leeftijd	bmi	actief
roken \					
count	200.000000	200.000000	200.000000	200.000000	200.000000
mean	100.500000	6.283400	61.895000	28.241310	1.955000
std	57.879185	0.802654	6.310554	2.961234	0.784892
min	1.000000	4.400000	49.000000	20.797000	1.000000
25%	50.750000	5.800000	57.750000	26.271250	1.000000
50%	100.500000	6.200000	62.000000	28.125000	2.000000
75%	150.250000	6.732500	65.000000	29.732500	3.000000

```
1.000000
max      200.000000      8.200000      84.000000      37.565000      3.000000
1.000000
```

```

      sekse      alcohol
count  200.000000  200.000000
mean    0.410000    0.765000
std     0.493068    0.722763
min     0.000000    0.000000
25%     0.000000    0.000000
50%     0.000000    1.000000
75%     1.000000    1.000000
max     1.000000    2.000000
```

```
array([0., 2., 1.])
```

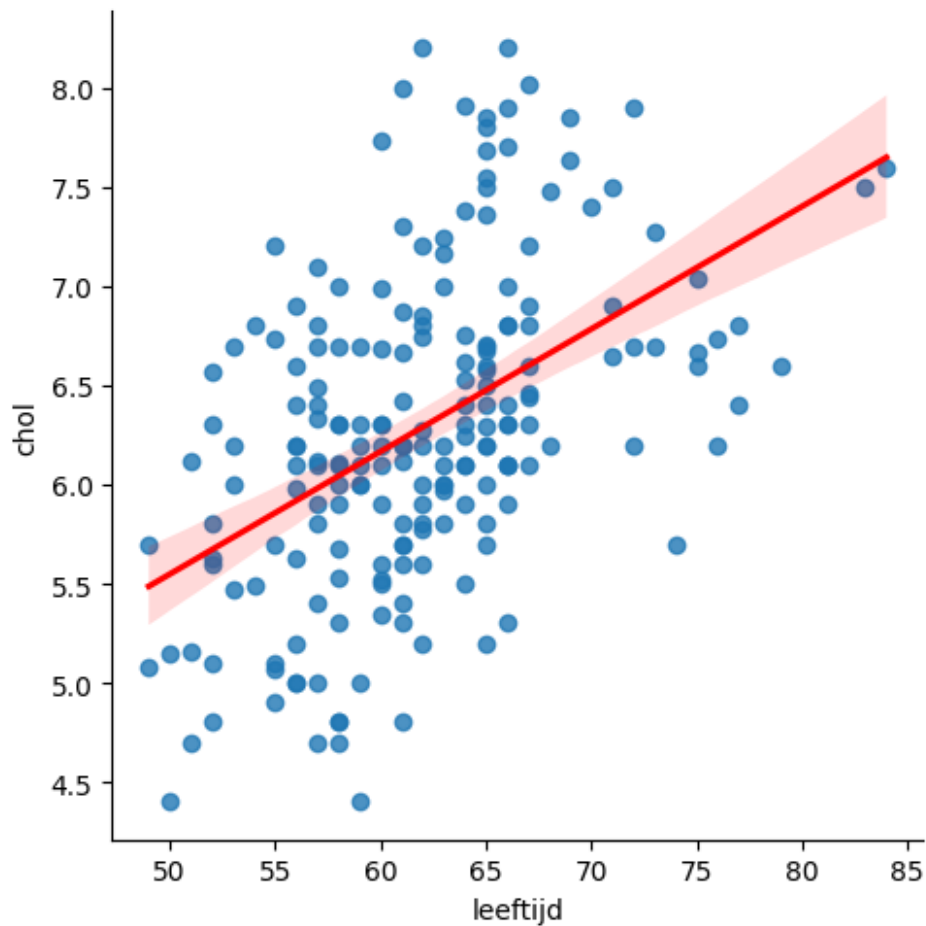
```
chol1
```

```

      id  chol  leeftijd      bmi  actief  roken  sekse  alcohol
0      1.0  7.90      66.0  27.410      1.0    0.0    1.0      0.0
1      2.0  7.50      83.0  26.850      1.0    0.0    1.0      2.0
2      3.0  7.60      84.0  27.240      1.0    0.0    1.0      2.0
3      4.0  6.00      65.0  27.500      1.0    0.0    1.0      2.0
4      5.0  6.20      56.0  27.780      1.0    0.0    1.0      0.0
..      ...      ...      ...      ...      ...      ...      ...
195  196.0  6.53      64.0  30.417      3.0    1.0    0.0      1.0
196  197.0  6.50      65.0  30.300      3.0    1.0    0.0      1.0
197  198.0  6.40      66.0  29.910      3.0    1.0    0.0      1.0
198  199.0  5.50      64.0  27.400      3.0    1.0    0.0      1.0
199  200.0  6.90      71.0  33.860      3.0    1.0    0.0      1.0
```

```
[200 rows x 8 columns]
```

```
sns.lmplot(data=chol1, y='chol', x='leeftijd', fit_reg=True,
line_kws={"color": "red"})
plt.show()
```



INTERPRETATION: As age increases, cholesterol levels tend to rise. However, the spread of points around the line indicates that other factors may also influence cholesterol levels. This plot suggests that age is a significant but not exclusive predictor of cholesterol levels.

(b) Fit a Linear Model

Fit a linear model for chol with leeftijd using the function `ols` (using the `statsmodels` library). The formula for the model is `chol ~ leeftijd`. Save the fitobject under the name `fit1`. View the result with `fit1.summary()`.

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

fit1 = smf.ols("chol ~ leeftijd", data=chol1).fit()
print(fit1.summary())
```

OLS Regression Results

```
=====
=====
Dep. Variable:          chol    R-squared:
0.236
```

```

Model:                                OLS    Adj. R-squared:
0.232
Method:                               Least Squares    F-statistic:
61.19
Date:                                Thu, 20 Feb 2025    Prob (F-statistic):
3.04e-13
Time:                                15:44:13    Log-Likelihood:
-212.39
No. Observations:                    200    AIC:
428.8
Df Residuals:                        198    BIC:
435.4
Df Model:                            1

Covariance Type:                    nonrobust

=====
=====
              coef      std err          t      P>|t|      [0.025
0.975]
-----
-----
Intercept      2.4584      0.492      5.002      0.000      1.489
3.428
leeftijd       0.0618      0.008      7.822      0.000      0.046
0.077
=====
=====
Omnibus:              3.098    Durbin-Watson:
1.036
Prob(Omnibus):        0.212    Jarque-Bera (JB):
3.162
Skew:                 0.286    Prob(JB):
0.206
Kurtosis:             2.770    Cond. No.
615.
=====
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.

```

INTERPRETATION: The model indicates a significant positive relationship between age and cholesterol levels. However, with an R-squared of only 0.236, age alone does not explain the majority of the variation in cholesterol levels. Adding other relevant variables (such as smoking status, gender, education level, etc.) may improve the model's performance.

(c) Fit a Multiple Linear Model

Fit a model fit2 for chol with leeftijd, bmi, sekse and alcohol. Which factors are statistically significant?

```
fit2 = smf.ols("chol ~ leeftijd + bmi + sekse + alcohol",  
data=choll).fit()  
print(fit2.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          chol    R-squared:
0.396
Model:                  OLS     Adj. R-squared:
0.383
Method:                 Least Squares    F-statistic:
31.93
Date:                  Thu, 20 Feb 2025    Prob (F-statistic):
1.83e-20
Time:                  15:44:13    Log-Likelihood:
-188.94
No. Observations:      200    AIC:
387.9
Df Residuals:          195    BIC:
404.4
Df Model:               4
```

Covariance Type: nonrobust

```
=====
=====
              coef      std err          t      P>|t|      [0.025
0.975]
-----
-----
Intercept    -0.1829      0.609     -0.301      0.764     -1.383
1.017
leeftijd      0.0329      0.008      3.921      0.000      0.016
0.049
bmi           0.1330      0.019      6.937      0.000      0.095
0.171
sekse         1.0012      0.189      5.293      0.000      0.628
1.374
alcohol       0.3492      0.109      3.212      0.002      0.135
0.564
```

```
=====
=====
Omnibus:      4.439    Durbin-Watson:
```

```

1.015
Prob(Omnibus):                0.109   Jarque-Bera (JB):
4.221
Skew:                        0.258   Prob(JB):
0.121
Kurtosis:                    3.489   Cond. No.
940.
=====
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

INTERPRETATION: The model demonstrates that age, BMI, sex, and alcohol consumption are all significant predictors of cholesterol levels. Including these additional variables improved the model's explanatory power from 23.6% to 39.6%.

COMMENT

Considering fit2, we see that the all the factors are statistically significant beside the Intercept. As their p-values are less than 0.05.

```

## this if we want to use one-hot encoding
### convert to factors
chol2 = chol1.copy()
chol2["sekse"] = chol2["sekse"].astype(str)
chol2["alcohol"] = chol2["alcohol"].astype(str)

chol2 = pd.get_dummies(chol2, columns=["sekse", "alcohol"],
drop_first=True)

chol2 = chol2.rename(columns={
    "sekse_1.0": "sekse_1",
    "alcohol_1.0": "alcohol_Low",
    "alcohol_2.0": "alcohol_High"
})

print(chol2.columns
)
fit2_factors = smf.ols("chol ~ leeftijd + bmi + actief + roken +
sekse_1 + alcohol_Low + alcohol_High", data=chol2).fit()
print(fit2_factors.summary())

Index(['id', 'chol', 'leeftijd', 'bmi', 'actief', 'roken', 'sekse_1',
      'alcohol_Low', 'alcohol_High'],
      dtype='object')

```

OLS Regression Results

```

=====
=====
Dep. Variable:                chol    R-squared:
0.489
Model:                        OLS     Adj. R-squared:
0.471
Method:                      Least Squares    F-statistic:
26.29
Date:                        Thu, 20 Feb 2025    Prob (F-statistic):
4.64e-25
Time:                        15:44:13    Log-Likelihood:
-172.11
No. Observations:            200    AIC:
360.2
Df Residuals:                192    BIC:
386.6
Df Model:                    7

Covariance Type:            nonrobust

=====
=====
                                coef    std err          t      P>|t|
-----
-----
Intercept                    -0.7894    0.575      -1.372    0.172
-1.924    0.345
sekse_1[T.True]              0.5087    0.264       1.927    0.055
-0.012    1.029
alcohol_Low[T.True]         -0.4757    0.269      -1.770    0.078
-1.006    0.055
alcohol_High[T.True]        0.1643    0.249       0.661    0.510
-0.326    0.655
leeftijd                     0.0354    0.008       4.220    0.000
0.019    0.052
bmi                          0.1672    0.020       8.297    0.000
0.127    0.207
actief                      -0.0044    0.063      -0.070    0.945
-0.128    0.120
roken                       0.3861    0.100       3.870    0.000
0.189    0.583

=====
=====
Omnibus:                     6.605    Durbin-Watson:
1.136
Prob(Omnibus):               0.037    Jarque-Bera (JB):
6.780
Skew:                        0.448    Prob(JB):

```

0.0337

Kurtosis:

2.899

Cond. No.

965.

=====

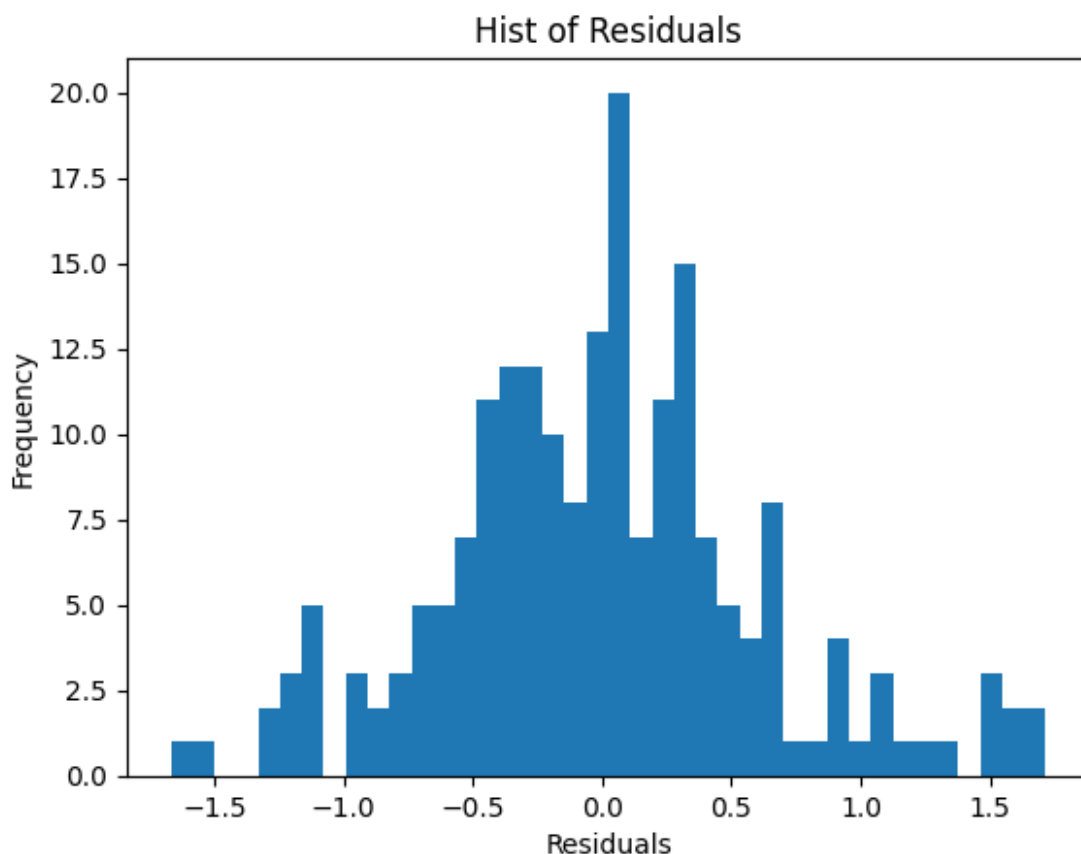
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

INTERPRETATION: The model shows that Age, BMI, and Smoking are strong predictors of cholesterol levels. However, some variables like Physical Activity, Alcohol Consumption, and Gender are not clearly significant. The R-squared of 0.489 is an improvement from previous models, but there may still be unexplained variance.

(d) Residuals and Histogram

```
choll["residuals"] = fit2.resid
plt.hist(choll["residuals"], bins=40)
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.title("Hist of Residuals")
plt.show()
```



INTERPRETATION: The residuals are approximately normally distributed, which supports the validity of the regression model. Minor deviations from normality are not likely to affect the model's performance significantly.

2.7 DATASET: Births

```
births = pd.read_csv('births-1.csv')
births['home'] = births['child_birth'].apply(lambda x: 'at_home' if
'first line' in x and 'at home' in x else 'not_at_home')
```

```
births.head(10)
```

	provmin	urban	child_birth
0	68	strong	first line child birth, at home
1	12	moderate	first line child birth, outpatient
2	99	not	first line child birth, outpatient
3	68	moderate	during pregnancy referred to specialist
4	68	strong	originally at home, referred to specialist
5	11	not	first line child birth, at home
6	69	strong	first line child birth, at home
7	99	not	originally outpatient, referred to specialist
8	12	not	originally at home, referred to specialist
9	68	very strong	first line child birth, at home

	age_cat	age	ethnicity	parity	home
0	25-29 year	26	Dutch	2	at_home
1	25-29 year	29	Dutch	1	not_at_home
2	25-29 year	25	Mediterranean	2	not_at_home
3	30-34 year	30	Dutch	3	not_at_home
4	30-34 year	33	Dutch	1	not_at_home
5	> 35 year	39	Dutch	3	at_home
6	30-34 year	32	Dutch	4	at_home
7	25-29 year	25	Dutch	3	not_at_home
8	25-29 year	29	Dutch	1	not_at_home
9	25-29 year	29	Dutch	2	at_home

(b) Recode the variable parity in a new variable pari where pari has level primi if it concerns a first

```
births['pari'] = births['parity'].apply(lambda x: 'primi' if x<2 else 'multi')
```

(c) Recode the variable ethnicity into a new variable etni where etni has level Dutch if the woman

```
births['ethnicity'].unique()

array(['Dutch', 'Mediterranean', 'Hindu', 'other European', 'Creole',
       'Asian', 'other'], dtype=object)

births['etni'] = births['ethnicity'].apply(lambda x: 'Dutch' if
x=='Dutch' else 'Not Dutch')
```

(d) Logistic Regression Model

- Using the sklearn library, create a logistic regression model with the function `LogisticRegression` to predict the probability of childbirth at home.
- Use the variables:
 - `pari`, `age_cat` (age categorized), `etni`, and `urban` (urbanization degree).
- View the model's outcomes using:

```
classification_report()
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import OneHotEncoder
from sklearn.metrics import classification_report
from sklearn.model_selection import train_test_split

# One-Hot Encoding categorical features
categorical_features = ['age_cat', 'etni', 'urban']
encoder = OneHotEncoder(drop='first', sparse=False)
encoded_features = encoder.fit_transform(births[categorical_features])
encoded_df = pd.DataFrame(encoded_features,
columns=encoder.get_feature_names_out(categorical_features))

# Merge with numerical data
X = pd.concat([births[['parity']], encoded_df], axis=1)
y = births['home']

# Convert target variable to binary (if needed)
y = y.map({'at_home': 1, 'not_at_home': 0}) # Adjust based on actual labels

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=7)
```

```

clf = LogisticRegression(random_state=0)
clf.fit(X_train, y_train)

y_pred = clf.predict(X_test)

report = classification_report(y_test, y_pred)
print(report)

```

	precision	recall	f1-score	support
0	0.65	0.83	0.73	5866
1	0.59	0.35	0.44	4075
accuracy			0.63	9941
macro avg	0.62	0.59	0.58	9941
weighted avg	0.63	0.63	0.61	9941

```

c:\Users\Mardeen\AppData\Local\Programs\Python\Python311\Lib\site-
packages\sklearn\preprocessing\_encoders.py:972: FutureWarning:
`sparse` was renamed to `sparse_output` in version 1.2 and will be
removed in 1.4. `sparse_output` is ignored unless you leave `sparse`
to its default value.
  warnings.warn(

```

INTERPRETATION: The model performs reasonably well for predicting class 0 but struggles with class 1, particularly due to low recall. This suggests the model has a high false negative rate for class 1, which could be problematic depending on the application. The imbalance between precision and recall for class 1 indicates that the model may need further optimization or rebalancing techniques.

(e) Decision Tree Model

```

from sklearn.tree import DecisionTreeClassifier, plot_tree
dt_model = DecisionTreeClassifier(random_state=0, max_depth=4)
dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)

report = classification_report(y_test, y_pred)
print("Decision Tree Classification Report:\n", report)

# Plot Decision Tree
plt.figure(figsize=(15, 8))
plot_tree(dt_model, feature_names=X.columns.tolist(),
class_names=['not_at_home', 'at_home'], filled=True)
plt.show()

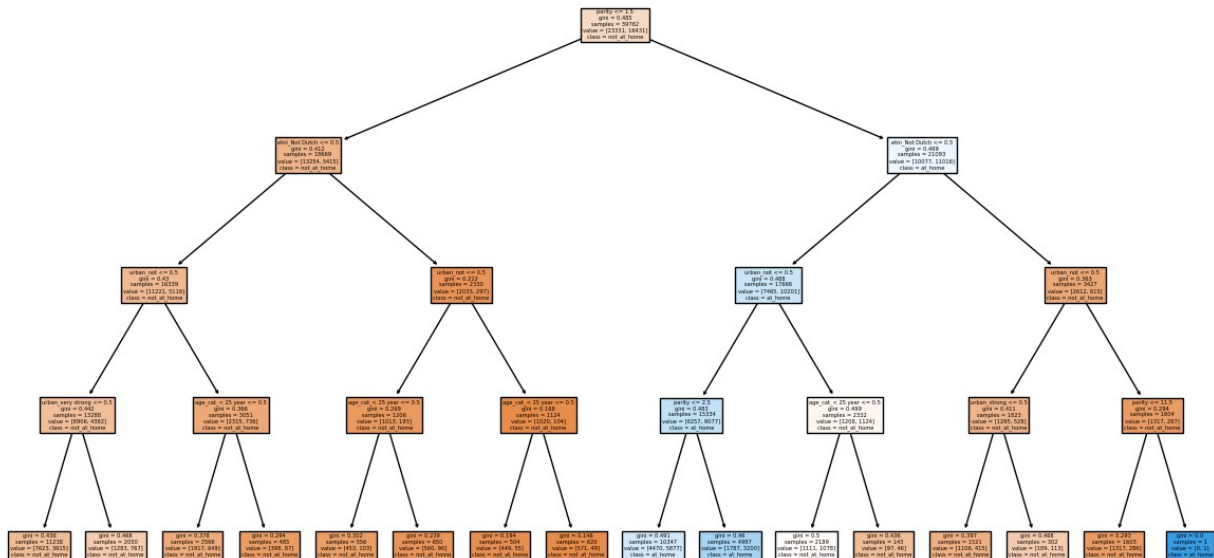
```

```

Decision Tree Classification Report:
precision    recall  f1-score   support

```

	0	0.69	0.72	0.71	5866
	1	0.58	0.54	0.56	4075
accuracy				0.65	9941
macro avg		0.64	0.63	0.63	9941
weighted avg		0.65	0.65	0.65	9941



INTERPRETATION: The Decision Tree model performs better for class 0 than for class 1, especially in terms of recall. While the overall accuracy is acceptable, the lower recall and precision for class 1 suggest that the model struggles with identifying positive instances correctly.

(f) Model Comparison with Cross-Validation

```
from sklearn.model_selection import cross_val_score

log_reg = LogisticRegression(random_state=0)
decision_tree = DecisionTreeClassifier(random_state=0, max_depth=4)

# Perform Cross-Validation (5-Fold)
log_reg_scores = cross_val_score(log_reg, X_train, y_train, cv=5,
scoring='accuracy')
dt_scores = cross_val_score(decision_tree, X_train, y_train, cv=5,
scoring='accuracy')

print(f"Logistic Regression Accuracy (Cross-Validation Mean):
{log_reg_scores.mean():.4f}")
print(f"Decision Tree Accuracy (Cross-Validation Mean):
{dt_scores.mean():.4f}")
```

```
better_model = "Logistic Regression" if log_reg_scores.mean() >
dt_scores.mean() else "Decision Tree"
print(f"\nBetter Model based on Cross-Validation: {better_model}")
```

Logistic Regression Accuracy (Cross-Validation Mean): 0.6424

Decision Tree Accuracy (Cross-Validation Mean): 0.6571

Better Model based on Cross-Validation: Decision Tree

INTERPRETATION: Based on Cross-Validation Accuracy, the Decision Tree model is the better-performing model. However, it is important to consider other performance metrics like precision, recall, F1-score, and ROC-AUC, especially if the dataset is imbalanced or if there are concerns about false positives or false negatives.